

Mathematical Specification: Spatial CRT Design Simulation with Heterogeneous Incidence

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1 Introduction

This document provides a rigorous mathematical specification for the simulation study evaluating treatment assignment designs in Spatial Cluster Randomized Trials (CRTs) under heterogeneous outcome incidence. The study investigates which sampling design strategy yields the most efficient (lowest error) estimates of the direct intervention effect τ when spillover effects and spatial dependence are present.

Research question: Given known historical incidence data exhibiting spatial heterogeneity, which treatment assignment design minimizes estimation error for τ across a range of spatial dependence and spillover configurations?

Application context: Sudden Unexpected Death (SUD) in North Carolina, where county-level and census tract-level incidence rates are modeled via Poisson regression (Gan et al., 2019; Watson et al., AHA Abstract). The base SUD incidence rate is approximately 35 per 100,000 (Mirzaei et al., 2019).

2 Study Region and Spatial Structure

2.1 Regular Lattice Grid

The study region is a regular square lattice grid:

$$\mathcal{G} = \{s_1, s_2, \dots, s_N\}$$

where each cluster s_i is located at integer coordinates (x_i, y_i) on a $G \times G$ grid, so $N = G^2$. In this study, $G = 10$ and $N = 100$.

2.2 Spatial Contiguity

Two contiguity definitions are considered:

- **Rook contiguity:** clusters sharing an edge (4-connected). The neighbor set is

$$\mathcal{N}_R(s_i) = \{s_j : |x_i - x_j| + |y_i - y_j| = 1\}$$

- **Queen contiguity:** clusters sharing an edge or vertex (8-connected). The neighbor set is

$$\mathcal{N}_Q(s_i) = \{s_j : \max(|x_i - x_j|, |y_i - y_j|) = 1\}$$

2.3 Spatial Weight Matrix

Let $\mathbf{A}$ denote the binary adjacency matrix where $A_{ij} = \mathbf{1}[s_j \in \mathcal{N}(s_i)]$. The row-standardized spatial weight matrix is:

$$W_{ij} = \frac{A_{ij}}{\sum_{k=1}^N A_{ik}}, \quad i \neq j; \quad W_{ii} = 0$$

This ensures $\sum_j W_{ij} = 1$ for each row, so spatial lags represent neighborhood averages. We construct $\mathbf{W}_R$ (rook) and $\mathbf{W}_Q$ (queen) weight matrices.

3 Outcome Data Generating Process

3.1 Spatial Lag Model with a Spillover Term

The outcome vector $\mathbf{Y} = (Y_1, \dots, Y_N)^\top$ is generated from a spatial lag model with a spillover term (called the ‘‘Spatial Durbin Model’’, SDM, in older project documents):

$$\mathbf{Y} = (\mathbf{I}_N - \rho \mathbf{W})^{-1} [\tau \mathbf{Z} + S(\mathbf{Z}) + \beta \mathbf{X} + \boldsymbol{\varepsilon}], \quad \beta = 1, \sigma = 1$$

where:

Symbol	Description	Role
$\mathbf{Y}$	$N \times 1$ outcome vector	Response
$\mathbf{I}_N$	$N \times N$ identity matrix	—
ρ	Spatial autocorrelation parameter	Controls global spatial dependence in $\mathbf{Y}$
$\mathbf{W}$	$N \times N$ row-standardized weight matrix	Spatial structure (rook or queen)
τ	Direct treatment effect	Estimand of interest
$\mathbf{Z}$	$N \times 1$ binary treatment vector	$Z_i \in \{0, 1\}$
γ	Spillover effect magnitude	Indirect effect via treated neighbors
$S(\mathbf{Z})$	$N \times 1$ spillover term (includes γ)	See Section 3.2
β	Effect of baseline incidence on outcome	Covariate coefficient
$\mathbf{X}$	$N \times 1$ baseline incidence vector	$X_i \in [0, 1]$
$\boldsymbol{\varepsilon}$	$N \times 1$ error vector	$\varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$

The spatial multiplier $(\mathbf{I}_N - \rho \mathbf{W})^{-1}$ induces global spatial dependence: each cluster’s outcome depends on the entire network through higher-order neighbors.

Stationarity condition: The model requires $|\rho| < 1/\lambda_{\max}(\mathbf{W})$. For row-standardized $\mathbf{W}$, $\lambda_{\max} = 1$, so we require $|\rho| < 1$.

3.2 Spillover Construction

The spillover term captures indirect treatment effects transmitted through the spatial network. It is built from the spatial lag of treatment,

$$(\mathbf{WZ})_i = \frac{\sum_{j \in \mathcal{N}(i)} Z_j}{|\mathcal{N}(i)|},$$

the proportion of cluster i 's neighbors that are treated. Two spillover regimes are considered; in both, $S(\mathbf{Z})$ already contains the magnitude γ :

1. **Control-only spillover** (`control_only`): the spillover effect applies only to untreated clusters:

$$S_i^{(\text{ctrl})}(\mathbf{Z}) = \gamma \cdot (\mathbf{WZ})_i \cdot (1 - Z_i), \quad \text{i.e. } S(\mathbf{Z}) = \gamma \mathbf{WZ} \circ (\mathbf{1} - \mathbf{Z})$$

2. **Bidirectional spillover** (`both`): the spillover effect applies to all clusters:

$$S_i^{(\text{both})}(\mathbf{Z}) = \gamma \cdot (\mathbf{WZ})_i, \quad \text{i.e. } S(\mathbf{Z}) = \gamma \mathbf{WZ}$$

The oracle estimator (Section 6.2) uses $\text{Spill} = S(\mathbf{Z})$, the true regime-specific spillover term, as a covariate.

4 Incidence Generation Modes

The baseline incidence vector $\mathbf{X}$ represents prior observed outcome rates across clusters. We consider three generation modes with increasing structural complexity. All modes produce values in $[0, 1]$ to ensure comparability across modes and with the treatment assignment designs.

Each incidence configuration $c = (\text{mode}, \rho_X)$ yields $K = 10$ independent incidence surfaces $\mathbf{X}_1, \dots, \mathbf{X}_K$ (Section 7.2). The same surface $\mathbf{X}_k$ is used to draw the treatment assignments, to generate the outcome, and as the covariate in the analysis model. The spatial filters in Modes 2 and 3 always use the queen weight matrix $\mathbf{W}_Q$, whichever contiguity is used for the outcome.

4.1 Mode 1: Independent and Identically Distributed (iid)

$$X_i \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1), \quad i = 1, \dots, N$$

This serves as the **naïve baseline** with no spatial structure in incidence. It represents the case where historical outcome rates show no geographic clustering.

Properties: $\mathbb{E}[X_i] = 0.5$, $\text{Var}(X_i) = 1/12$, $\text{Cor}(X_i, X_j) = 0$ for $i \neq j$.

4.2 Mode 2: Spatially Correlated (SAR Filter)

Spatially correlated incidence is generated via the Simultaneous Autoregressive (SAR) filter:

$$\mathbf{X}^* = (\mathbf{I}_N - \rho_X \mathbf{W}_Q)^{-1} \mathbf{u}, \quad \mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_N)$$

where $\rho_X \in [0, 1)$ controls the degree of spatial correlation in incidence.

The raw values X_i^* are then transformed to $[0, 1]$ via the probability integral transform:

$$X_i = \Phi(X_i^*)$$

where $\Phi(\cdot)$ is the standard normal CDF.

Key property: When $\rho_X = 0$, $\mathbf{X}^* = \mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_N)$ and $X_i = \Phi(u_i) \sim \text{Uniform}(0, 1)$ by the probability integral transform, recovering Mode 1 marginally. When $\rho_X > 0$, the X_i remain marginally approximately uniform but exhibit positive spatial correlation: nearby clusters have similar incidence values.

Parameters: $\rho_X \in \{0.20, 0.50\}$ (incidence spatial correlation).

4.3 Mode 3: Poisson Count-Based

Motivated by the SUD application, this mode generates incidence from a Poisson count model with spatially correlated rates:

Step 1. Generate spatially correlated log-rates:

$$\log \boldsymbol{\lambda}^* = \mu_0 \cdot \mathbf{1} + (\mathbf{I}_N - \rho_X \mathbf{W}_Q)^{-1} \mathbf{u}, \quad \mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_N)$$

where $\mu_0 = \log(\text{base rate})$ is the log of the baseline incidence rate (e.g., $\mu_0 = \log(35/100,000)$ for SUD).

Step 2. Generate Poisson counts given cluster populations:

$$C_i \sim \text{Poisson}(\lambda_i^* \cdot P_i)$$

where P_i is the population of cluster i .

Step 3. Compute observed rates:

$$R_i = C_i / P_i$$

Step 4. Rank-normalize to $[0, 1]$:

$$X_i = \frac{\text{rank}(R_i) - 0.5}{N}$$

This quantile-based transformation preserves the ordering of rates while mapping to a uniform marginal distribution on $[0, 1]$.

With $P_0 = 100,000$ people per cluster, the expected count is $\lambda_i^* P_0 \approx 35$ deaths per cluster, so the observed rates R_i carry real rank information. (Before the 2026-09 revision, $P_0 = 1,000$ gave $\lambda_i^* P_0 \approx 0.35$ expected deaths and only about five distinct values of X_i .) Any remaining ties in $\mathbf{X}_k$ are broken at random by the designs (Section 5).

4.3.1 Population Modes

Two population specifications are available:

1. **Equal population:** $P_i = P_0$ for all i (default: $P_0 = 100,000$); used in the simulation (`pop_mode = "equal"`)
2. **Heterogeneous population:** $P_i \sim \text{LogNormal}(\log P_0, \sigma_P^2)$, where $\sigma_P = 0.5$ provides moderate population heterogeneity (available in the code, not used in the simulation)

Parameters: $\rho_X \in \{0.20, 0.50\}$, base rate = 35/100,000, $P_0 = 100,000$, population mode = equal.

5 Treatment Assignment Designs

Eight designs are evaluated. Each design produces a binary treatment vector $\mathbf{Z} \in \{0, 1\}^N$. For surface k and design draw j , the assignment is $\mathbf{Z}_{kj} = \text{design}_d(\mathbf{X}_k; \text{draw } j)$: designs that use incidence see the same surface $\mathbf{X}_k$ that enters the outcome. Checkerboard (Design 1) is the only deterministic design; Designs 2–8 involve randomization.

Tie-breaking. Ties in $\mathbf{X}_k$ are broken at random, freshly for every design draw j , using the random-tie rank

$$r_i = \text{rank}(X_{k,i}), \quad \text{ties.method} = \text{"random"},$$

so $(r_1, \dots, r_N)$ is a permutation of $1, \dots, N$. Designs 2, 6, 7 and 8 use r (or random-tie ranks of quadrant means); the other designs do not use incidence.

5.1 Design 1: Checkerboard (Strict Spatial Stratification)

$$Z_i = (x_i + y_i) \mod 2$$

Treatment is assigned in an alternating checkerboard pattern. Every edge-neighbor of a treated cluster is a control and vice versa, so the two arms are maximally *interspersed*: the design maximizes treated-control adjacency. It does not separate the arms; that is the Isolation Buffer’s aim.

Properties: Deterministic (the same $\mathbf{Z}$ for every k and j ; each fit still has its own noise column $\boldsymbol{\varepsilon}_{kj}$), exactly 50 treated. $\mathbf{Z}$ is perfectly negatively autocorrelated under rook contiguity.

Near-collinearity under queen contiguity. Each interior cluster has 4 edge-neighbors in the other arm and 4 corner-neighbors in its own arm, so $(\mathbf{W}_Q \mathbf{Z})_i = 1/2$ for every interior cluster whatever its own assignment. $\mathbf{W}_Q \mathbf{Z}$ varies only along the grid boundary (range 1/3 to 2/3, SD 0.066), which leaves the spillover term little variation independent of the intercept and $\mathbf{Z}$. The estimate stays unbiased, but its variance is large.

Aliasing under rook contiguity. On the rook grid every neighbor of a treated cluster is a control and vice versa, so $\mathbf{W}_R \mathbf{Z} = \mathbf{1} - \mathbf{Z}$ and $\text{rank}([\mathbf{1}, \mathbf{Z}, \mathbf{W}_R \mathbf{Z}]) < 3$. The oracle model’s Spill column is then aliased with the intercept and $\mathbf{Z}$ and is dropped, so τ is not identified apart from γ . These estimates are kept and flagged (Section 8.7).

5.2 Design 2: High Incidence Focus

$$Z_i = \mathbf{1}[r_i > N/2]$$

Treatment is assigned to the $N/2$ clusters with the highest historical incidence, with ties broken at random as above. This represents a strategy that deliberately targets high-burden areas.

Properties: Randomized only through the tie-break (a fixed assignment when $\mathbf{X}_k$ has no ties), exactly 50 treated, treatment correlated with $\mathbf{X}$.

5.3 Design 3: Saturation Quadrants

The grid is partitioned into four spatial quadrants:

$$Q_\ell, \quad \ell = 1, 2, 3, 4$$

defined by dividing at the grid midpoint $(G/2, G/2)$, so each quadrant is a 5×5 block with $|Q_\ell| = 25$. For each randomization:

1. Randomly assign saturation levels $\{0.20, 0.40, 0.60, 0.80\}$ to the four quadrants (a random permutation, one per quadrant)
2. Within quadrant Q_ℓ with saturation p_ℓ , randomly select $\lfloor |Q_\ell| \cdot p_\ell \rfloor$ clusters for treatment

Properties: Stochastic, treatment proportion varies by region (5, 10, 15 and 20 of 25), captures varying treatment density; exactly 50 treated overall.

5.4 Design 4: Isolation Buffer

Treatment is assigned sequentially with spatial constraints:

1. Initialize all clusters as available
2. Randomly select an available cluster and assign treatment ($Z_i = 1$)
3. Remove the selected cluster and all its neighbors from the available set
4. Repeat until no available clusters remain

Properties: Stochastic (a greedy random maximal independent set in the active neighbor graph), treated clusters are never adjacent (under the active contiguity), creates a spatial buffer around each treated unit. The number treated varies across realizations: about 38 under rook and about 22 under queen contiguity.

Under rook contiguity a draw can occasionally equal one colour class of the bipartite rook grid, i.e. the exact checkerboard, and is then aliased in the same way as Design 1 (Section 8.7).

5.5 Design 5: 2x2 Blocking

The grid is partitioned into non-overlapping 2×2 blocks:

$$B_m = \{s_i : \lfloor (x_i - 1)/2 \rfloor = a_m, \lfloor (y_i - 1)/2 \rfloor = b_m\}, \quad |B_m| = 4$$

Within each block, exactly 2 of 4 clusters are randomly assigned to treatment:

$$\sum_{i \in B_m} Z_i = 2 \quad \text{for all } m$$

Properties: Stochastic, exactly 50% treated, locally balanced (1:1 within each block).

5.6 Design 6: Balanced Quartiles

Clusters are stratified into four equal-size quartiles of the random-tie rank r (`ntile(r, 4)`):

$$S_q = \{s_i : (q - 1)N/4 < r_i \leq qN/4\}, \quad |S_q| = 25, \quad q = 1, 2, 3, 4$$

Each stratum treats $\lfloor 25/2 \rfloor = 12$ clusters, and two randomly chosen strata treat one more, so that exactly $N/2$ are treated:

$$\sum_{i \in S_q} Z_i \in \{12, 13\}, \quad \sum_{q=1}^4 \sum_{i \in S_q} Z_i = 50$$

Properties: Stochastic, exactly 50 treated, balanced across the incidence distribution. Combines covariate stratification with randomization. (Before the 2026-09 revision it treated $4 \times 12 = 48$, from `round(25/2) = 12` per stratum.)

5.7 Design 7: Balanced Halves

Clusters are stratified into two equal-size halves of the random-tie rank r (`ntile(r, 2)`):

$$S_1 = \{s_i : r_i > N/2\} \text{ (high incidence)}, \quad S_2 = \{s_i : r_i \leq N/2\} \text{ (low incidence)}, \quad |S_h| = 50$$

Within each half, exactly half the clusters are randomly assigned to treatment:

$$\sum_{i \in S_h} Z_i = |S_h|/2 = 25, \quad h = 1, 2$$

Properties: Stochastic assignment. Exactly 50 treated overall. Balanced across the two halves of the incidence distribution. This is a 2-stratum analog of Design 6 (Balanced Quartiles), which uses 4 strata.

5.8 Design 8: Incidence-Guided Saturation Quadrants

The grid is partitioned into four spatial quadrants $Q_1, \dots, Q_4$ using the same spatial partition as Design 3. Let $\bar{X}_{Q_\ell} = |Q_\ell|^{-1} \sum_{i \in Q_\ell} X_i$ denote the average baseline incidence in quadrant ℓ .

Quadrants are ranked by $\bar{X}_{Q_\ell}$ in descending order (ties between quadrant means broken at random, freshly for every design draw) and assigned saturation levels by rank:

Rank (by $\bar{X}_{Q_\ell}$)	Saturation p_ℓ
1 (highest)	0.80
2	0.60
3	0.40
4 (lowest)	0.20

Within each quadrant, $\lfloor |Q_\ell| \cdot p_\ell \rfloor$ clusters are randomly selected for treatment:

$$\sum_{i \in Q_\ell} Z_i = \lfloor |Q_\ell| \cdot p_\ell \rfloor$$

Properties: The quadrant-to-saturation mapping is fixed by the incidence surface $\mathbf{X}_k$ (up to random tie-breaking between equal quadrant means); within-quadrant assignment is stochastic. Exactly 50 treated overall. Unlike Design 3, the saturation assignment is guided by incidence data rather than random.

6 Estimation Methods

6.1 Difference in Means (DIM)

The naive estimator:

$$\hat{\tau}_{\text{DIM}} = \bar{Y}_1 - \bar{Y}_0 = \frac{\sum_{i:Z_i=1} Y_i}{\sum_i Z_i} - \frac{\sum_{i:Z_i=0} Y_i}{\sum_i (1 - Z_i)}$$

Variance estimation (Neyman):

$$\widehat{\text{Var}}(\hat{\tau}_{\text{DIM}}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0}$$

where S_g^2 is the sample variance in group g and $n_g = \sum_i \mathbf{1}[Z_i = g]$.

Confidence interval:

$$\hat{\tau}_{\text{DIM}} \pm z_{0.975} \cdot \sqrt{\widehat{\text{Var}}(\hat{\tau}_{\text{DIM}})}$$

Note: DIM is biased for τ when spillover ($\gamma \neq 0$) or spatial dependence ($\rho \neq 0$) is present, because it ignores the spatial structure of the DGP.

Status: DIM is a naive baseline only and is **not** part of the revised (2026-09) simulation; `05_run_simulation.R` fits the ML estimators below. The existing DIM results are pre-revision (old DGP, 6 designs) and are not comparable to the revised ML results.

6.2 Maximum Likelihood Estimation (MLE)

The spatial lag model is estimated by maximum likelihood. For fit (k, j) the oracle model is

$$Y_i = \rho \sum_{i'} W_{ii'} Y_{i'} + \alpha + \tau Z_i + \delta \cdot \text{Spill}_i + \beta X_{k,i} + \varepsilon_i$$

or in matrix form:

$$\mathbf{Y} = \rho \mathbf{W} \mathbf{Y} + \mathbf{X}_{\text{design}} \boldsymbol{\beta}_{\text{design}} + \boldsymbol{\varepsilon}$$

where $\mathbf{X}_{\text{design}} = [\mathbf{1}, \mathbf{Z}, \text{Spill}, \mathbf{X}_k]$ and $\boldsymbol{\beta}_{\text{design}} = (\alpha, \tau, \delta, \beta)^\top$. Because $\text{Spill} = S(\mathbf{Z})$ already contains γ (Section 3.2), its true coefficient is $\delta = 1$, and the true intercept is $\alpha = 0$. The analysis uses the same incidence surface $\mathbf{X}_k$ that generated the outcome, and the same $\mathbf{W}$ (rook or queen) as the DGP.

The MLE maximizes the concentrated log-likelihood in ρ ,

$$\ell(\rho) = \log |\mathbf{I}_N - \rho \mathbf{W}| - \frac{N}{2} \log(2\pi) - \frac{N}{2} \log \hat{\sigma}^2(\rho) - \frac{N}{2}, \quad \log |\mathbf{I}_N - \rho \mathbf{W}| = \sum_{i=1}^N \log(1 - \rho \lambda_i),$$

over $\rho \in (1/\lambda_{\min}(\mathbf{W}), 1/\lambda_{\max}(\mathbf{W}))$ (the interval $1/\text{range}(\text{eig}(\mathbf{W}))$, shrunk by machine ϵ), where λ_i are the eigenvalues of $\mathbf{W}$, $\hat{\sigma}^2(\rho) = \|\mathbf{e}_0 - \rho \mathbf{e}_W\|^2/N$, and $\mathbf{e}_0, \mathbf{e}_W$ are the OLS residuals of $\mathbf{Y}$ and of $\mathbf{W}\mathbf{Y}$ on $\mathbf{X}_{\text{design}}$. Given $\hat{\rho}$, $\hat{\boldsymbol{\beta}}_{\text{design}}$ is the OLS fit of $\mathbf{Y} - \hat{\rho} \mathbf{W}\mathbf{Y}$ on $\mathbf{X}_{\text{design}}$. Standard errors come from inverting the analytic information matrix for $(\sigma^2, \rho, \boldsymbol{\beta}_{\text{design}})$, scaled by s^2 . The 95% Wald confidence interval for τ is

$$[L, U] = \hat{\tau}_{\text{MLE}} \mp z_{0.975} \cdot \text{SE}(\hat{\tau}_{\text{MLE}}), \quad z_{0.975} = 1.96.$$

Engine. The fits use `fit_sar_lag()` (code/04_estimation.R), a lean implementation of the same estimator as `spatialreg::lagsarlm(method = "eigen")`: the eigenvalues of $\mathbf{W}$ are computed once per neighbor type rather than once per fit. It was validated against `lagsarlm` on 5,120 fits (`results/estimator_validation/`): maximum $|\Delta \hat{\tau}| = 9.7\text{e-}8$, relative $\Delta \text{SE} = 1.1\text{e-}8$, identical aliasing, 0.6 vs 76 ms per fit (almost all of `lagsarlm`'s time is a `gc()` call on exit). `engine = "lagsarlm"` is retained for a 1% cross-check.

If a column of $\mathbf{X}_{\text{design}}$ is aliased (linearly dependent on the others; same QR rule as `lm`, tolerance 10^{-7}), the engine warns “Aliased variables found”, drops that column, and fits the rest. Aliased estimates are kept and flagged (Section 8.7).

6.2.1 Oracle vs. Non-Oracle Estimation

Two estimators are fit to the same simulated outcome $\mathbf{Y}_{kj}$; both include an intercept:

- **Oracle (primary):** $Y \sim Z + \text{Spill} + X_k$, with $\text{Spill} = S(\mathbf{Z}_{kj})$, the true regime-specific spillover term. This provides the best-case estimation performance.
- **Non-oracle (sensitivity analysis):** $Y \sim Z + X_k$. The model omits the spillover term, reflecting what a researcher would estimate without knowledge of the spillover mechanism.

Because both use the same $\mathbf{Y}$, they share every random draw (Section 7.5). Results are stored in separate files, one per estimator.

7 Simulation Algorithm

7.1 Parameter Space

The simulation sweeps over the following parameter grid:

Parameter	Symbol	Values	Count
Incidence mode	—	iid, spatial, poisson	3 modes
Incidence spatial correlation	ρ_X	0 (iid), 0.20, 0.50	5 configs
Neighbor type	—	rook, queen	2
Spatial autocorrelation	ρ	0.00, 0.01, 0.20, 0.50	4
Spillover magnitude	γ	0.50, 0.60, 0.70, 0.80	4
Spillover type	—	control_only, both	2
Treatment design	—	1, 2, 3, 4, 5, 6, 7, 8	8
Treatment effect	τ	0.8, 1.0, 1.5, 2.0, 3.0	5

Total scenarios: 5 incidence configs $\times$ 2 neighbor types $\times$ 4 ρ $\times$ 4 γ $\times$ 2 spillover types $\times$ 8 designs $\times$ 5 τ = **12,800 scenarios**, each fit by both the oracle and the non-oracle estimator.

Notation: incidence config $c = (\text{mode}, \rho_X)$; block $b = (c, \text{nb}, \rho, \gamma, \text{regime})$; scenario $s = (b, d, \tau)$ for design d .

7.2 Resample Structure

Each scenario is evaluated over a matched-surface Monte Carlo structure:

- K = incidence surfaces $\mathbf{X}_1, \dots, \mathbf{X}_K$ per config
- J = design draws per surface, $\mathbf{Z}_{kj} = \text{design}_d(\mathbf{X}_k; \text{draw } j)$
- One noise column $\boldsymbol{\varepsilon}_{kj} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_N)$ per fit
- Total fits per scenario: $K \times J$, i.e. $K \cdot J$ distinct $(\mathbf{Z}, \boldsymbol{\varepsilon})$ pairs over K surfaces

For fit (k, j) the outcome is generated from the same surface that the design and the analysis use:

$$\mathbf{Y}_{kj} = (\mathbf{I}_N - \rho \mathbf{W})^{-1} [\tau \mathbf{Z}_{kj} + S(\mathbf{Z}_{kj}) + \beta \mathbf{X}_k + \boldsymbol{\varepsilon}_{kj}]$$

Default values:

Estimation Mode	K (surfaces)	J (design draws)	Fits $K \cdot J$
MLE (oracle and non-oracle)	10	25	250
DIM (pre-revision)	—	—	$R_d \times R_o = 25 \times 100 = 2,500$

The DIM row describes the pre-revision runs (R_d design resamples $\times$ R_o outcome resamples), which used the old structure (below) and are not re-run in the revision.

Before the 2026-09 revision (April 2026 runs, for reference): the $K = 10$ surfaces were drawn once per (c, τ) ; the designs saw only $\mathbf{X}_1$, while fit (j, k) used $\mathbf{X}_k$ in the outcome and analysis, so $\text{cor}(\mathbf{X}_1, \mathbf{X}_k) \approx 0$ and the design and the outcome saw unrelated incidence surfaces. The 10 noise columns were drawn once per (c, nb, ρ) and each was reused by all 25 design draws, and by every γ , regime and design.

7.3 Pseudocode

```

BUILD spatial_grid -> coords, W_rook, W_queen, listw_rook, listw_queen
PRECOMPUTE lag_setup (eigenvalues of W, rho interval) for rook and queen
SET I_mat = diag(N)

```

```

FOR EACH config c = (incidence_mode, rho_X) IN {iid; spatial 0.20, 0.50; poisson 0.20, 0.50}:
  FOR k IN 1:K:

```

```

SEED(key = "X", mode, rho_X, k)
GENERATE surface X_k (length N) using incidence_mode, rho_X, W_queen
# X_1..X_K are shared by every block of config c

FOR EACH nb_type IN {rook, queen}:          # work unit = (c, nb_type, rho)
  FOR EACH rho IN {0.00, 0.01, 0.20, 0.50}:
    SELECT active W for nb_type
    COMPUTE inv_mat = solve(I_mat - rho * W)

  FOR EACH gamma IN {0.50, 0.60, 0.70, 0.80}:
    FOR EACH spill_type IN {control_only, both}:    # block b
      SEED(key = "eps", mode, rho_X, nb_type, rho, gamma, spill_type)
      GENERATE E (N x K*J) ~ N(0, sigma^2)    # one column per fit, shared by all
                                              # designs, taus and both estimators

    FOR EACH design_id IN {1, ..., 8}:
      FOR k IN 1:K:
        SEED(key = "Z", mode, rho_X, nb_type, rho, gamma, spill_type, k, design_id)
        GENERATE J assignments Z_kj = design(X_k)    # random tie-breaking
        COMPUTE Spill_kj = S(Z_kj) (gamma * W Z, masked by (1 - Z) if control_only)
        FLAG Z_WZ_rank_deficient = any rank([1, Z_kj, W Z_kj]) < 3

      FOR EACH tau IN {0.8, 1.0, 1.5, 2.0, 3.0}:    # same Z and E for every tau
        Y_kj = inv_mat %*% (tau * Z_kj + Spill_kj + beta * X_k + E[, kj])
        FIT oracle      Y_kj ~ Z + Spill + X_k    (fit_sar_lag; warnings captured)
        FIT non-oracle Y_kj ~ Z + X_k
        FOR EACH estimator:
          COMPUTE Bias, SD, MSE, Coverage, Power, Fail_Rate, N_Valid_Est
          COMPUTE surface-level metrics m_1..m_K and MC SEs
          COUNT N_Aliased, N_Warn
          STORE scenario row and K surface rows

```

Work units (c, nb, ρ) are run in parallel; because every draw is seeded from its key, the output does not depend on the order or the number of workers. If any unit fails, the run stops.

Critical structural invariant: Each incidence surface $\mathbf{X}_k$ is generated **once per (incidence_mode, ρ_X) configuration** and held fixed across all spatial parameter combinations (ρ , neighbor type, γ , regime), designs and τ . Only the assignments, the noise and the outcome DGP vary across blocks. This reflects the real-world scenario where historical incidence is observed once, and the question is how different spatial models and designs affect estimation quality.

7.4 Deterministic Design Optimization

Design 1 (Checkerboard) is the only deterministic design: it produces the same treatment vector for every surface k and draw j , so a single assignment is replicated. Each of its $K \cdot J$ fits still has its own noise column ε_{kj} , so it contributes 250 distinct fits like every other design. (Before the revision, High Incidence Focus was also treated as deterministic; it is now drawn with random tie-breaking.)

7.5 Per-Scenario Seeding

Rather than a single global seed, every random draw is preceded by

```

set.seed(digest::digest2int(key), kind = "Mersenne-Twister",
        normal.kind = "Inversion", sample.kind = "Rejection")

```

where the key joins the fields below with "|", numeric fields formatted with `sprintf("%.2f")`:

Draw	Key fields
$\mathbf{X}_k$ (one surface)	"X", mode, ρ_X , k
$\mathbf{Z}_k$. (all J draws for surface k)	"Z", mode, ρ_X , nb, ρ , γ , regime, k , d
ε (all 250 columns for a block)	"eps", mode, ρ_X , nb, ρ , γ , regime

Consequences:

- Each draw depends only on its key, so results are identical sequentially, in parallel, and in any order, and do not change if scenarios are added, removed, or reordered.
- $\mathbf{X}_k$ is shared by every block of config c .
- **Common random numbers:** within a block, all 8 designs share ε and $\mathbf{X}_k$; each τ reuses the same $\mathbf{Z}$ and ε ; the oracle and non-oracle fits use the same $\mathbf{Y}$.
- $\mathbf{Z}$ and ε are independent across blocks, but blocks within a config share the $\mathbf{X}$ surfaces.

7.6 Estimand and Dependence Structure

For scenario s , the target is performance **averaged over incidence surfaces**:

$$\text{MSE}(s) = \mathbb{E}_{\mathbf{X}} \mathbb{E}_{\mathbf{Z}|\mathbf{X}} \mathbb{E}_{\varepsilon} [(\hat{\tau} - \tau)^2],$$

and likewise for bias, coverage and power. $\mathbf{X}$ is drawn from the config's generator, $\mathbf{Z}$ from the design given $\mathbf{X}$, and $\varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_N)$.

Within a scenario: $K = 10$ independent surfaces; given $\mathbf{X}_k$, $J = 25$ independent $(\mathbf{Z}_{kj}, \varepsilon_{kj})$ pairs. The surface is the primary sampling unit, so Monte Carlo SEs are computed from the $K = 10$ surface-level means (Section 8.8), not from 250 fits treated as independent.

Across scenarios:

- **Designs within a block** are paired through common ε and $\mathbf{X}_k$ (a within-block comparison removes that shared noise).
- τ levels share $\mathbf{Z}$ and ε . Per- τ Friedman tests are therefore not independent confirmations of each other.
- **Blocks within a config** share $\mathbf{X}_1, \dots, \mathbf{X}_{10}$. A Friedman test over blocks treats them as independent, which overstates the information about surface-to-surface variation. The surface-level robustness check addresses this: for the key pairwise claims it runs a paired test over 50 independent (config $\times$ k) units, 5 configs $\times$ 10 surfaces, with the metric for each unit averaged over that surface's blocks.

8 Evaluation Metrics

For each scenario and estimator, let $\hat{\tau}_1, \dots, \hat{\tau}_n$ denote the estimates of τ from the n valid fits (finite $\hat{\tau}$ and finite SE) among the $K \cdot J = 250$ fits, indexed by f , with Wald interval $[L_f, U_f]$ from Section 6.2.

8.1 Bias

$$\text{Bias} = \bar{\hat{\tau}} - \tau, \quad \text{where } \bar{\hat{\tau}} = \frac{1}{n} \sum_{f=1}^n \hat{\tau}_f$$

8.2 Standard Deviation

$$\text{SD} = \sqrt{\frac{1}{n-1} \sum_{f=1}^n (\hat{\tau}_f - \bar{\hat{\tau}})^2}$$

8.3 Mean Squared Error

$$\text{MSE} = \frac{1}{n} \sum_{f=1}^n (\hat{\tau}_f - \tau)^2 = \text{Bias}^2 + \text{Var}(\hat{\tau})$$

MSE decomposes into squared bias plus variance, capturing both accuracy and precision.

8.4 Coverage Probability

$$\text{Coverage} = \frac{1}{n} \sum_{f=1}^n \mathbf{1}[L_f \leq \tau \leq U_f]$$

where $[L_f, U_f] = \hat{\tau}_f \mp 1.96 \cdot \text{SE}_f$ is the 95% confidence interval for the f -th estimate. Nominal coverage is 0.95; under- or over-coverage indicates CI miscalibration.

8.5 Power

$$\text{Power} = \frac{1}{n} \sum_{f=1}^n \mathbf{1}[L_f > 0]$$

the proportion of fits that reject $H_0 : \tau = 0$. Power is one-sided (the interval lies entirely above zero), as in the pre-revision runs.

8.6 Convergence Failure Rate

$$\text{Fail Rate} = 1 - \frac{n}{K \cdot J}, \quad \text{N_Valid_Est} = n$$

This captures the proportion of fits with a non-finite $\hat{\tau}$ or SE (an engine error, or an information matrix that cannot be inverted). Collinearity between $\mathbf{Z}$ and Spill does **not** produce a failure: the aliased column is dropped and the fit is kept and flagged (Section 8.7).

8.7 Aliasing and Warning Flags

Each scenario row records:

- **N_Aliased**: the number of fits where the engine warned “Aliased variables found”.
- **N_Warn**: the number of all other warnings (every message is written to the warning log).
- **Z_WZ_rank_deficient**: TRUE if $\text{rank}([\mathbf{1}, \mathbf{Z}, \mathbf{WZ}]) < 3$ for any $\mathbf{Z}$ draw of the scenario. It is recorded in both estimator files; the non-oracle model has no Spill column to alias, but its Checkerboard $\times$ rook estimates share the same cause (τ is not identified apart from γ).

Aliased estimates are **kept** and flagged in exhibits. Besides Checkerboard $\times$ rook (every fit; Section 5), Isolation Buffer $\times$ rook can occasionally draw the exact checkerboard: a maximal independent set that equals one colour class of the bipartite rook grid. In the full run that happened in 4 of its 40,000 rook draws (4 blocks $\times$ 5 $\tau = 20$ scenarios, 1 aliased fit each). An oracle fit is aliased exactly when its draw is rank-deficient.

8.8 Monte Carlo Standard Errors

Monte Carlo SEs are computed at the surface level (Section 7.6). For each metric m (bias, MSE, coverage, power), compute m_k on surface k ’s valid fits, $k = 1, \dots, K$; then

$$\text{SE}(m) = \frac{\text{sd}(m_1, \dots, m_K)}{\sqrt{K}}, \quad m \pm t_{0.975, K-1} \cdot \text{SE}(m) \quad (t_9 \text{ for } K = 10).$$

The SE is taken over the surfaces with at least one valid fit (**N_Surfaces**). Design differences are paired at the surface level: the SE of a difference is that of $(m_{d,k} - m_{d',k})$ over k . The per-surface metrics are saved in the surface-level results file (one row per scenario and surface).

9 Complete Parameter Table

Parameter	Symbol	Default	Sweep Values	Configurable
Grid dimension	G	10	—	Yes
Number of clusters	N	100	G^2	Derived
True treatment effect	τ	1.0 (primary)	0.8, 1.0, 1.5, 2.0, 3.0	Yes
Incidence coefficient	β	1.0	—	Yes
Error std. deviation	σ	1.0	—	Yes
Spatial autocorrelation	ρ	—	0.00, 0.01, 0.20, 0.50	Yes
Spillover magnitude	γ	—	0.50, 0.60, 0.70, 0.80	Yes
Spillover type	—	—	control_only, both	Yes
Neighbor type	—	—	rook, queen	Yes
Incidence mode	—	—	iid, spatial, poisson	Yes
Incidence spatial corr.	ρ_X	—	0 (iid), 0.20, 0.50	Yes
Base rate (Poisson)	μ_0	35/100,000	—	Yes
Cluster population	P_0	100,000	—	Yes
Population mode	—	equal	equal (heterogeneous available, not used)	Yes
Incidence surfaces per config (MLE)	K	10	—	Yes
Design draws per surface (MLE)	J	25	—	Yes
Fits per scenario (MLE)	$K \cdot J$	250	—	Derived
Design resamples (DIM, pre-revision)	R_d	25	—	Yes
Outcome resamples (DIM, pre-revision)	R_o	100	—	Yes
Design IDs	—	{1,2,3,4,5,6,7,8}	—	Yes

Parameter	Symbol	Default	Sweep Values	Configurable
Estimators	—	oracle (primary)	oracle, non-oracle (both fit to the same Y)	—
ML engine	—	<code>fit_sar_lag()</code>	<code>fit_sar_lag()</code> , <code>lagsarlm</code> (1% cross-check)	Yes
Estimation mode	—	MLE	MLE (DIM pre-revision only)	Yes

All parameters are configurable to allow investigators to apply these findings to their own study contexts, even when some parameters (e.g., spillover magnitude) would not be known in practice.

10 References

- Gan, C. et al. (2019). County-level SUD incidence in North Carolina via Poisson regression.
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- LeSage, J. & Pace, R.K. (2009). *Introduction to Spatial Econometrics*. CRC Press.
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